Ethical Lens: Justice and Fairness

Justice & Fairness

AI Loan Approval

- Let’s start off with a relevant AI example.

- An AI loan system uses historical data to decide who gets approved. In the past, some neighborhoods—mostly home to families of color—were unfairly labeled as risky because of racial bias. The AI learns that pattern and still treats people from those areas as high risk today.

Predictive Policing

- Here’s another relevant example.

- Police use predictive software that disproportionately targets neighborhoods with high populations of people of color. As a result, more patrols are sent there which creates a feedback loop: over-policing leads to more arrests; in turn, this outcome gives the appearance of reinforcing the validity of the biased data.

Justice & Fairness

As these examples show, algorithms can replicate and deepen historical injustices. Fairness in AI, therefore, must account for disparate impacts, systemic bias, and the need for equitable access and treatment across populations.

Defining Justice and Fairness

Justice

- Justice means giving each person what they deserve or their due. It’s often associated with a standard of rightness and has been a cornerstone of ethical discussions throughout history.

Fairness

- Fairness is closely related to justice but often refers two things: (1) the ability to judge without bias (not allowing feelings or personal interests to cloud your judgment) and (2) making judgments that are appropriate to specific cases (and not mistaking them for more general circumstances).

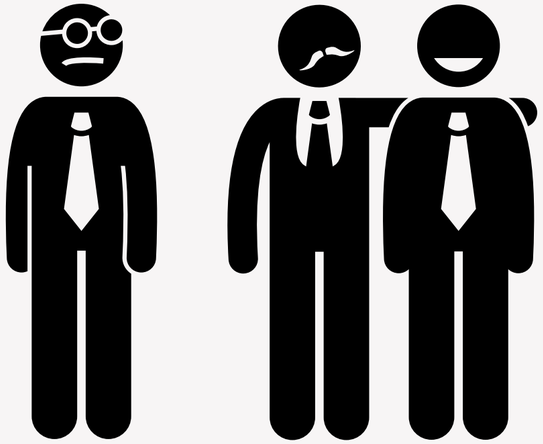
The Fundamental Principle of Justice

- “Equals should be treated equally and unequals unequally.” - Aristotle

- This principle, first defined by Aristotle more than 2000 years ago, forms the foundation of modern justice.

- It suggests that individuals should be treated the same unless there are relevant differences between them. For instance, equal work should result in equal pay, regardless of gender or race.

Justifiable Criteria for Differential Treatment



Need

- The government provides benefits to the needy that it doesn’t give to affluent citizens.

Desert

- The government punishes those who have broken the law, not those who haven’t.

Contribution

- Rewarding those who contribute more to a project with greater benefits.

Effort

- Recognizing and rewarding those who exert more effort in a project.

Unjustifiable Criteria for Differential Treatment

Age

- Discriminating based on age in work environments is generally considered unjust.

Sex

- Treating individuals differently based on their sex is typically seen as unfair.

Race

- Racial discrimination is widely recognized as an unjust basis for differential treatment.

Religious Preferences

- Using religious beliefs as a basis for unequal treatment is generally deemed unjust.

Different Types of Justice

Distributive Justice

- Fair distribution of benefits and burdens in society.

Retributive / Corrective Justice

- Fair and just punishments, for example, punishments that consider the severity of the crime and the intent of the criminal, but discount irrelevant criteria, such as race.

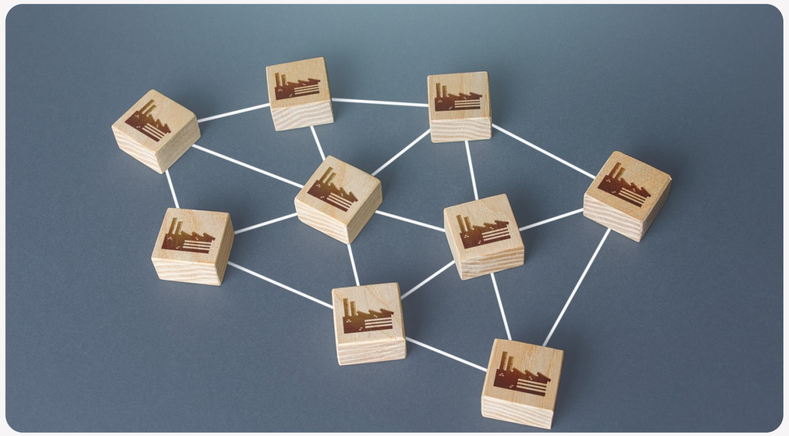
Compensatory Justice

- Fair compensation for injuries by those who have inflicted the harm—for example, coal mine companies paying workers who got sick doing their jobs.

Foundations of Justice

Social Stability

- Justice ensures societal stability by promoting fair treatment.
- When people feel they've been treated unfairly, they may engage in activities that cause social unrest.



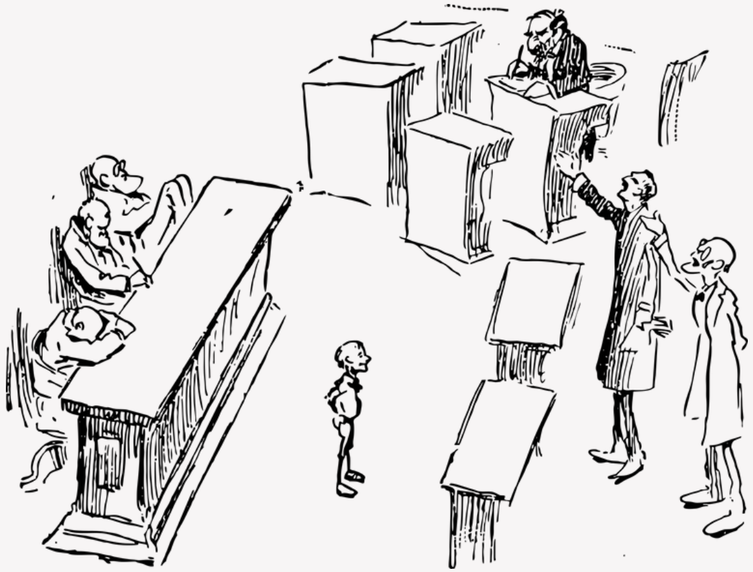
Interdependence

- Just institutions maintain social unity in interdependent communities.
- Since societies require cooperation and divided labor, people who feel they're treated unfairly can stop giving their best contributions.

Equal Dignity

- Recognition of inherent human dignity demands equal treatment.
- People who feel their fundamental dignity is violated can respond in anti-social ways.

Justice in Ethical Decision-Making



Evaluate Equal Treatment

- Consider whether actions treat all persons equally.

Justify Differences

- If unequal treatment exists, determine if the criteria are relevant and justified.

Balance with Other Principles

- Consider other moral claims like rights or societal welfare.